Q5

Show that the roots of the following quadratic eqs are rational.

i $((l-m)^2 x^2 + (m+n-l)x - n = 0$

$(l-m)x^2 + (m+n-l)x - n = 0$

$a = (l-m)$

$b = (m+n-l)$

$c = -n$

$\Delta = b^2 - 4ac$

$\Delta = (m+n-l)^2 - 4(l-m)(-n)$

we KNow that $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ac$

we can write like this

$\Delta = (\underset{a}{m}+\underset{b}{n}+(\underset{c}{-l}))^2 - 4(-ln+mn)$

$\Delta = (m+n+(-l))^2 - 4(l-m)(-n)$

$\Delta = (m+n+(-l))^2 - 4(-nl+mn)$

$\Delta = (m+n+(-l))^2 + 4ln - 4mn$

$\Delta = (m)^2 + (n)^2 + (-l)^2 + 2(m)(n) + 2(n)(-l) + 2(m)(-l) + 4ln - 4mn$

$\Delta = m^2 - n^2 + l^2 + \cancel{2mn} - 2nl - 2ml + \cancel{4mL} - 4mn$

$\Delta = m^2 - n^2 + l^2 - 2mn + 2ln - 2mL$

$\Delta = m^2 - n^2 + L^2 - 2(mn + ln - mL)$

Now give - symbol inpoont of the m in this way formula will properly happen.

$$\Delta = (-m)^2 + (n)^2 (l)^2 + 2(-m)(n) + 2(n)(l) + 2(-m)(l)$$

$$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ac$$

Now Pack this formula: $(a+b+c)^2$

$$(-m+n+l)^2$$

Ans:

So we have to make the perfect square hence it has been done. and proved that quadratic equation are rational.